Offsets in Regression

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Introduction

In count regression, an offset is a predictor whose coefficient is fixed at 1. It is the standard mechanism for modelling rates per unit exposure: events per person-year of follow-up, infections per 1000 hospital admissions, mutations per kilobase of DNA. Without an offset, the model fits expected counts and treats exposure as a confounder; with an offset, the model fits expected rates and exposure is correctly accounted for. Forgetting the offset is a leading cause of biased coefficient estimates in epidemiological count-data work.

Prerequisites

A working understanding of Poisson and negative-binomial regression, the log link, and the distinction between counts and rates as inferential targets.

Theory

For events Y_i observed over exposure E_i, modelling the rate Y_i/E_i via Poisson regression is equivalent to modelling the count Y_i with $\log E_i$ as a known offset:

$$\log \mathbb{E}[Y_i] = \log E_i + X_i^\top \beta \quad \Leftrightarrow \quad \log \mathbb{E}[Y_i/E_i] = X_i^\top \beta.$$

The offset $\log E_i$ enters the linear predictor with a fixed coefficient of 1, so its contribution is treated as known rather than estimated. The remaining coefficients β are interpretable as log-rate ratios; exponentiating gives rate ratios per unit covariate change.

The same construction extends to negative-binomial regression (with overdispersion), to logistic regression with `offset = log(odds_baseline)` for known baseline rates, and to Cox regression where person-time at risk plays an analogous role.

Assumptions

Exposure $E_i > 0$ for every observation; the rate is proportional to exposure (a doubling of exposure doubles the expected count, all else equal); the count distribution (Poisson, NB) is appropriate for the data.

R Implementation

```r
set.seed(2026)
d <- data.frame(
  x = rnorm(200),
  person_years = runif(200, 0.5, 5)
)
d$events <- rpois(200, lambda = d$person_years * exp(-2 + 0.5 * d$x))

# With offset
fit <- glm(events ~ x, data = d, family = poisson,
           offset = log(person_years))
```

```r
summary(fit)
exp(coef(fit))      # rate ratios per unit x

# WITHOUT offset (wrong!): models expected count ignoring exposure
fit_no_off <- glm(events ~ x, data = d, family = poisson)
summary(fit_no_off)
```

Output & Results

With the offset, the slope coefficient is the log-rate ratio per unit increase in x; exponentiating gives the rate ratio (close to the data-generating value of $\exp(0.5) = 1.65$). Without the offset, coefficients absorb the variation in exposure and are biased; the comparison illustrates the importance of correctly specifying the rate model.

Interpretation

A reporting sentence: "After accounting for variable follow-up via an offset of log(person-years), each unit increase in x was associated with a 65 % higher event rate per person-year (rate ratio 1.65, 95 % CI 1.32 to 2.05, $p < 0.001$); omitting the offset gives a misleading 'count ratio' of 1.43 that conflates the rate effect with average follow-up." Always state the exposure unit (person-years, 1000 admissions, kilobases) when reporting rate ratios.

Practical Tips

- The offset is log(exposure), not exposure itself; using exposure directly is a common bug.
- The `offset = log(exposure)` argument in `glm()` keeps the coefficient fixed at 1; including `log(exposure)` as a regular predictor estimates the coefficient and produces a different model.
- Offsets generalise to negative-binomial, zero-inflated, and survival models; check the family-specific syntax (`glmmTMB`, `MASS::glm.nb`, `survival::coxph`).
- For binomial regression, the analogous construction uses `offset = log(p_baseline / (1 - p_baseline))` when a known baseline is to be respected.
- Always inspect whether exposure varies meaningfully across observations before deciding to include or omit an offset; ignoring a varying exposure biases all coefficient estimates.
- Report rate ratios after exponentiating; coefficient on the log scale is rarely the most communicative summary for clinical readers.

R Packages Used

Base R `glm()` with `offset =` and `family = poisson`; `MASS::glm.nb()` for negative-binomial regression with offsets; `glmmTMB` for mixed-effects count models with offsets and overdispersion; `survival::coxph()` for survival analyses where exposure enters as time at risk.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression

- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI